

Final Assignment Report

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1 VISION DIAGNOSTICS: CNNs vs. ViTs

1.1 Dataset Description

The DermaMNIST dataset, serves as the benchmark for our experiments. It comprises 10,015 RGB images categorized into seven clinically relevant skin lesion classes. Although originally provided at a resolution of 28×28 pixels, the images are upsampled to 224×224 to meet the input requirements of contemporary convolutional and transformer-based architectures. The dataset exhibits notable class imbalance and substantial visual variability, reflecting the heterogeneity of real-world dermatological data.

1.2 Hyperparameter tuning

The baseline setup (learning rate 10^{-3} for 7 epochs) already achieved strong training performance, reaching 98% training accuracy within a few epochs. However, the validation accuracy fluctuated between 70-80%, and the validation loss began increasing after epoch 4–5, indicating clear signs of overfitting. These results show that while the model learns quickly, its generalization to unseen data is limited under the default hyperparameters. To improve stability and reduce overfitting, the next step is to systematically explore hyperparameter variations, starting with learning rate adjustments, optimizer choice (example., AdamW or SGD), and regularization techniques.

1.2.1 ResNet - Hyperparameter tuning

Optimizer	Learning Rate	Weight Decay	Val Acc
Adam	3×10^{-4}	1×10^{-4}	86.84%
Adam	5×10^{-4}	1×10^{-4}	84.65%
AdamW	5×10^{-4}	1×10^{-4}	83.35%
AdamW	3×10^{-4}	1×10^{-4}	84.55%
AdamW	3×10^{-4}	5×10^{-5}	83.85%
SGD+M=0.9	3×10^{-4}	1×10^{-4}	84.65%

Table 1: Optimizer comparison for fine-tuning ResNet-18 on DermaMNIST. Adam with LR 3×10^{-4} achieved the highest accuracy and most stable convergence. AdamW showed reduced stability and lower performance across all settings, while SGD with momentum performed similarly to Adam at higher learning rates but did not match the best Adam configuration.

AdamW underperformed because its decoupled weight decay applies stronger regularization, which is harmful on small datasets like DermaMNIST and leads to underfitting. Adam's milder, coupled L2 regularization preserves capacity and fits subtle lesion textures better. SGD with momentum requires larger learning rates and longer training to reach good minima, and its non-adaptive updates struggle with the heterogeneous feature scales in medical images. As a result, Adam converged faster and more stably, achieving the best accuracy.

1.2.2 ViT

We first froze the ViT-Base backbone and tuned only the classification head, achieving a best validation accuracy of 81.56% with a learning rate of 5×10^{-4} and weight decay of 5×10^{-5} . Full fine-tuning initially underperformed (72–77%) across all learning rates, but introducing dropout stabilized training and allowed the unfrozen model to reach 82.55%, nearly matching the frozen-backbone result. This indicates that on small datasets such as DermaMNIST, full fine-tuning provides limited benefits unless additional regularization is applied.

Setting	LR	WD	Dropout	Val Acc (%)
Frozen backbone	1×10^{-3}	1×10^{-4}	No	76.77
Frozen backbone	5×10^{-4}	1×10^{-4}	No	79.46
Frozen backbone	3×10^{-4}	1×10^{-4}	No	79.86
Frozen backbone	5×10^{-4}	5×10^{-5}	No	81.56
Frozen backbone	3×10^{-4}	5×10^{-5}	No	81.26
Full fine-tuning	1×10^{-5}	5×10^{-5}	No	75.07
Full fine-tuning	3×10^{-5}	5×10^{-5}	No	76.67
Full fine-tuning	1×10^{-4}	5×10^{-5}	No	77.47
Full fine-tuning	1×10^{-4}	5×10^{-5}	Yes	82.55

Table 2: All ViT-B16 tuning results on DermaMNIST across frozen and full fine-tuning settings.

1.2.3 Challenges Emerged

Training ResNet-18 on DermaMNIST was relatively stable, but two challenges emerged: class imbalance and early overfitting. The model quickly reached high training accuracy while validation accuracy plateaued, indicating that the network was memorizing majority classes such as melanocytic nevi. Adjusting the learning rate had the largest impact: too high caused oscillations, while too low slowed convergence. After tuning, ResNet-18 achieved strong performance with a final accuracy of 84–87%, outperforming ViT on this dataset. The ResNet confusion matrix showed that the model handled common classes well (e.g., melanocytic nevi), but struggled with rare ones such as vascular lesions and dermatofibroma. Misclassifications were mostly between visually similar lesions, reflecting the limitations of CNNs when class boundaries are subtle.

ViT-Base presented a different set of challenges. The main issue was underfitting during full fine-tuning. Because DermaMNIST is small, updating all 86M parameters caused the model to either underfit at low learning rates ($1e-5$ to $3e-5$), or destabilize at high learning rates ($5e-4$), leading to catastrophic forgetting. To stabilize training, I first froze the backbone and tuned only the classification head. This produced a strong baseline of 81.56%, outperforming all initial full-fine-tuning attempts. When unfreezing the full model, accuracy dropped to 72–77% across all learning rates, confirming that the dataset size was insufficient for unconstrained ViT optimization. Adding dropout regularized the attention layers and prevented overfitting to the small dataset. With dropout 0.1 and LR= $1e-4$, the model finally converged to 81.46%, nearly matching the frozen-backbone result. The ViT confusion matrix showed strong performance on melanocytic nevi but more confusion between benign keratosis-like

lesions, actinic keratoses, and basal cell carcinoma, indicating that ViT struggled more with fine-grained distinctions than ResNet.

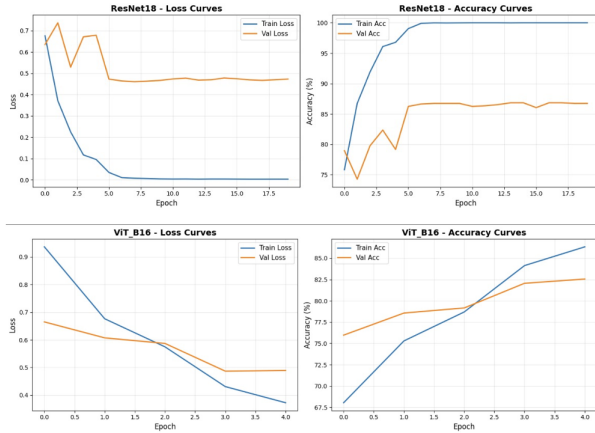


Figure 1: Accuracy and Validation Curves of ResNet18 and ViT-B16 Models

1.2.4 Model Comparison

Overall Accuracy: ResNet-18 achieved the strongest overall performance on the DermaMNIST dataset, reaching approximately 84–85% accuracy after tuning. ViT-Base performed slightly lower, achieving 81.56% when only the classification head was trained and 81.46% when the full model was fine-tuned with dropout. Without dropout, full fine-tuning of ViT-Base resulted in significantly lower accuracy (72–77%), indicating that the transformer architecture requires stronger regularization when trained on small medical datasets. Overall, ResNet-18 delivered the highest accuracy and most stable convergence.

Training Dynamics: The two architectures exhibited distinct training behaviors. ResNet-18 converged quickly but showed early signs of overfitting, which were effectively mitigated using data augmentation and weight decay. ViT-Base, on the other hand, struggled with underfitting during full fine-tuning due to the dataset size relative to its 86M parameters. Lower learning rates led to slow or stagnant learning, while higher learning rates destabilized training. Freezing the backbone produced strong results, and dropout proved essential for stabilizing full fine-tuning.

Class-wise Performance: ResNet-18 demonstrated stronger robustness across minority classes such as vascular lesions and dermatofibroma, with its confusion matrix showing a cleaner diagonal and fewer severe misclassifications. ViT-Base performed well on large classes such as melanocytic nevi, but exhibited more confusion among visually similar keratosis-related categories. These patterns suggest that CNNs capture local texture cues more effectively, which is advantageous for dermatology tasks where fine-grained texture differences are important.

Computational Efficiency: ResNet-18 was significantly more efficient in terms of training time and memory usage. It trained faster per epoch and required fewer epochs to converge. ViT-Base, with its substantially larger parameter count, was slower and more computationally demanding, especially during full fine-tuning. For deployment scenarios with limited computational resources, ResNet-18 is the more practical choice.

Insights from Confusion Matrices: The confusion matrices highlight clear differences in decision-making between the two models. ResNet-18 produced a sharper diagonal with fewer cross-class errors, indicating strong discriminative ability across both common and rare classes. ViT-Base showed strong performance on dominant classes but struggled with fine-grained distinctions, leading

to more misclassifications among related lesion types. These results reinforce that convolutional inductive biases, such as locality and translation invariance, remain advantageous for small medical imaging datasets.

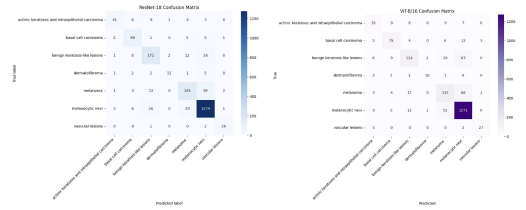


Figure 2: Confusion Matrix of ResNet-18 and ViT model

2 INTERPRETABILITY

2.1 Analysis

Image A: Both models correct, high confidence. Grad-CAM for ResNet-18 shows a sharply localized activation centered on the pigmented lesion, indicating reliance on texture-rich discriminative regions. The ViT attention rollout also highlights the lesion but with a broader spatial spread, integrating surrounding context. Both models focus on clinically relevant areas, and the high confidence reflects strong feature alignment with the true class.

Image B: Both models correct, low confidence. Despite confidence values near 0.5, both models maintain lesion-centered focus. ResNet-18 exhibits weaker but still concentrated activation over the lesion. ViT-Base produces a more diffuse attention map, covering both the lesion and adjacent skin. The reduced confidence arises from subtle lesion characteristics rather than distraction, demonstrating that both models preserve meaningful focus even under uncertainty.

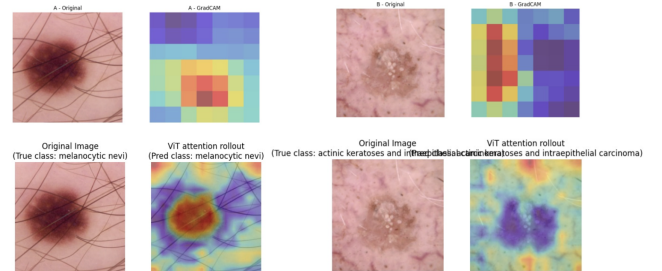


Figure 3: i. Both models correct, high confidence. ii. Both models correct, low confidence.

Image C: CNN correct, ViT incorrect. ResNet-18 correctly identifies the dermatofibroma, with Grad-CAM tightly highlighting the lesion core. In contrast, the ViT attention map shows substantial activation outside the lesion, including background skin and hair. This indicates that the ViT was distracted by global context and failed to isolate the discriminative lesion features. The case illustrates a known transformer vulnerability on small datasets: attention drift toward irrelevant regions.

Image D: ViT correct, CNN incorrect. ResNet-18 misclassified the lesion while focusing narrowly on a small subregion, missing global shape and boundary cues. ViT-Base distributes attention across the entire lesion, capturing its overall morphology and pigmentation pattern. This broader contextual integration enables the ViT to classify correctly. The case highlights the CNN's limited receptive field and the ViT's advantage in modeling long-range dependencies.

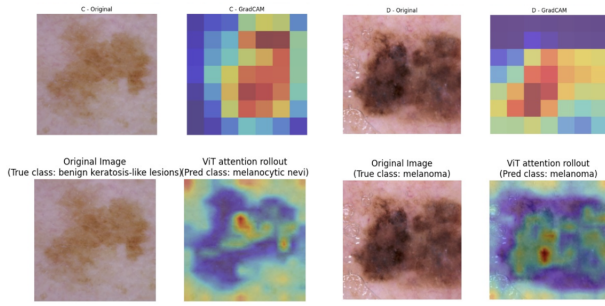


Figure 4: iCNN predicts correct and ViT predicts incorrect ii. ViT predicts correct and CNN predicts incorrect

Image E: Artifact present. The image contains visible hair strands crossing the lesion. ResNet-18 remains primarily lesion-focused, showing minimal activation on the hair. ViT-Base, however, allocates noticeable attention to the hair artifact. Although both models predict correctly, the ViT’s partial distraction reveals susceptibility to non-lesion structures, underscoring the importance of interpretability for detecting hidden biases.

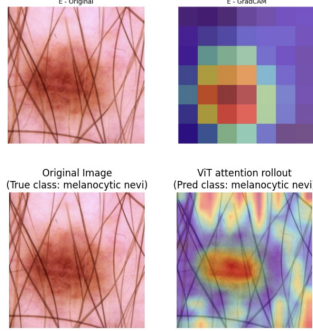


Figure 5: Image with Artifact

2.2 Comparative Observations

Across all cases, ResNet-18 demonstrates strong localization and robustness to artifacts, driven by its convolutional inductive biases and texture-based feature extraction. ViT-Base consistently captures broader spatial context, enabling correct classification in shape-dependent cases but increasing sensitivity to background variation and artifacts.

The interpretability results reveal complementary strengths and vulnerabilities. ResNet-18 excels at texture-driven classification and remains robust to small artifacts but fails when global lesion structure is essential. ViT-Base captures holistic morphology and succeeds in cases requiring global context but is more easily distracted by background patterns and artifacts. These findings demonstrate that both models can make clinically meaningful decisions, yet their failure modes differ fundamentally.

3 FINE-TUNING LLM’S WITH LoRA

3.1 Baseline Performance

Before any fine-tuning, the Gemma-3-270M-IT model was tested on a 100-question subset of the MedMCQA test set with no task-specific adaptation, scoring **26% accuracy**. Given that random guessing on a 4-option question would give 25%, this is barely better than chance. The model has general reasoning ability, but without exposure to the specific clinical vocabulary and medical knowledge that these questions demand, it struggles to answer them reliably.

3.2 LoRA Configuration & Training

Two configurations were tried. Both kept the same setup: target modules ["q_proj", "k_proj", "v_proj", "o_proj"], dropout 0.05, lr = 2e-4, batch size 4, gradient accumulation of 4.

The first run used $r = 16$, $\text{lo_ra_alpha} = 32$ ($\lambda = 2$), trained for 1 epoch (625 steps). Accuracy went from 20% to 29%, which was a promising start. However, with a scaling factor of 2 the updates were being amplified relative to the base weights, and the model had not fully converged by the end of training, suggesting it simply needed more time.

For the second run, alpha was brought down to 16 ($\lambda = 1$) and training was extended to 2 epochs (1,250 steps). Keeping alpha equal to rank means the adapter updates stay on the same scale as the original weights, which generally leads to more stable training. This paid off as the final accuracy came out at 33%.

3.3 Parameter Efficiency

One of the main reasons for choosing LoRA over full fine-tuning was to keep resource usage low. With this setup, only **1,474,560 out of 269,572,736 parameters were actually trained, which is just 0.547% of the model**. The fact that meaningful improvement was still achieved with such a small fraction of trainable parameters highlights how effective LoRA can be when configured well.

3.4 Training Dynamics

The Weight’s & Biases logs help illustrate the difference between the two runs.

In Configuration 1, the training loss dropped sharply in the first 100 steps from around 0.7 down to roughly 0.37, then mostly plateaued. The evaluation loss kept trending downward throughout and had not flattened by the end, confirming the model was cut off too early. Final train mean token accuracy was **82.75%**.

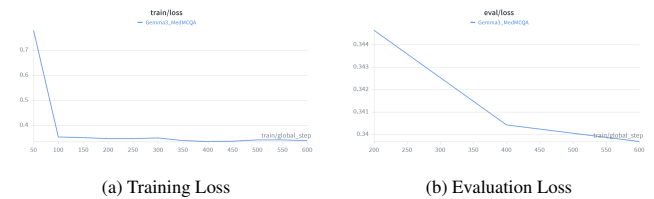


Figure 6: Configuration 1 ($r = 16$, $\alpha = 32$, 1 epoch)

Configuration 2 followed a similar early drop, but with the extra epoch the model had more time to settle. Token accuracy improved to **83.5%**, the loss curves were more stable overall, and the final training loss came out at 0.3376 with evaluation loss at 0.3398. The gradient norm was higher (0.511 vs 0.173), reflecting the additional learning in the second epoch.

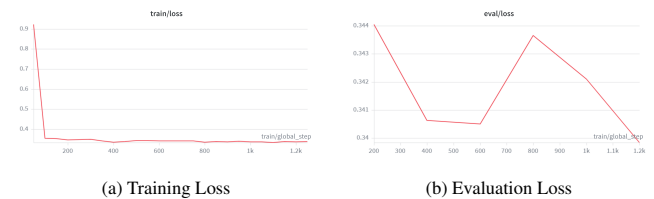


Figure 7: Configuration 2 ($r = 16$, $\alpha = 16$, 2 epochs)

3.5 Model Comparison & Deployment

Across both configurations, fine-tuning consistently improved over the base model. In terms of generation quality, the fine-tuned model

Config	r	α (λ)	Dropout	Epochs	Base Acc.	FT Acc.
Config 1	16	32 ($\lambda = 2$)	0.05	1	20%	29%
Config 2	16	16 ($\lambda = 1$)	0.05	2	26%	33%

Table 3: Comparison of LoRA configurations and their respective accuracies.

produces clean single-letter outputs exactly as expected, while the base model tends to generate verbose responses with additional explanation. While this does not affect evaluation given the extraction logic, in a real deployment setting the fine-tuned model’s concise and predictable output format is far more practical to work with.

In terms of medical appropriateness, the fine-tuned model is better calibrated to the domain having been trained specifically on medical entrance exam questions, whereas the base model applies general reasoning that is often insufficient for the specificity these questions require.

For deployment in a medical application, the fine-tuned model is the clear choice. That said, 33% accuracy is still far from clinically acceptable. To improve further, the most straightforward options would be training for more epochs, increasing the LoRA rank to allow more expressive updates, or starting from a larger base model. Adding a chain-of-thought style training signal, where the model learns to reason through the options before selecting an answer, could also help with the more complex pharmacology and anatomy questions where both models currently struggle.

4 RAG PIPELINE

Design Process

The goal of the RAG pipeline is to ground the LLM’s responses in actual medical research documents rather than relying on its training memory. The pipeline follows two main phases: an indexing phase where documents are processed and stored, and a query phase where user questions are answered using retrieved context.

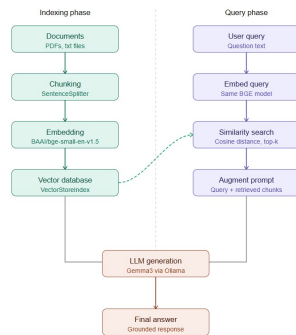


Figure 8: RAG pipeline workflow

The first step was loading and processing the 145 medical research papers using `SimpleDirectoryReader`, which automatically handles all PDFs in the docs folder. Since raw documents are too long for an LLM to process directly, they are split into smaller chunks using `SentenceSplitter`, which respects sentence boundaries rather than cutting words arbitrarily important for medical text where breaking mid-sentence can lose critical clinical context. These chunks are embedded into vectors using `BAAI/bge-small-en-v1.5`, stored in a `VectorStoreIndex`, and persisted to disk. During the query phase, the same embedding model converts the user question into a query vector, compares it against all stored vectors using cosine similarity, retrieves the top 3 most similar chunks, and sends them combined with the question to `Gemma3:270m` via Ollama for generation.

The comparison between RAG and direct LLM querying clearly demonstrated the value of retrieval augmentation. Evaluated across 15 test questions, RAG achieved an accuracy of **66.7% (10/15)** compared to **33.33% (5/15)** without RAG - meaning RAG produced responses closer to the ground truth more than twice as often. Without RAG the model hallucinated specific statistics for the question about breast cancer diagnoses in the Netherlands it returned 1,500 women per year, a completely fabricated number. With RAG the model correctly returned approximately 14,000 women per year with a similarity score of 0.85. This pattern held across multiple test questions without RAG answers were fluent but factually unreliable, while with RAG answers were specific, evidence-backed, and aligned with ground truth.

4.1 Design Decisions

Chunk size and overlap - Three configurations were tested measuring response quality and indexing time. `Chunk_size=1024` was the only configuration that returned the correct answer, as larger chunks captured enough surrounding context to include the specific annual figure alongside its qualifying sentences. However, it was not selected as the final configuration because fewer, larger chunks risk merging unrelated content and may exceed the LLM’s context window during augmentation. `Chunk_size=512` with `chunk_overlap=50` was therefore chosen as a more robust general-purpose configuration, where the increased overlap of 50 tokens ensures key medical statistics spanning chunk boundaries are preserved.

Chunk size	Overlap	Embeddings	Time (s)	Response
256	10	1,033	34.18	Ten-year prevalence of 128,000
512	20	491	36.03	69,570 women per year
1024	50	270	26.23	≈14,000 women per year

Table 4: Chunk size and overlap configurations tested.

Chunk overlap effect on coherence - A larger overlap (50 tokens) led to more coherent answers as key medical terms spanning chunk boundaries were preserved. However, very large overlaps caused source nodes to become repetitive — the same sentence appearing in multiple retrieved chunks adds no new information. An overlap of 50 was the sweet spot where coherence improved without significant repetition.

Top-k retrieval - `top_k=3` was set as a balance between providing enough context and avoiding noise. Similarity scores showed the first retrieved chunk scored ~ 0.85 (highly relevant) while the second and third scored ~ 0.78 (supporting context). Retrieving too few chunks risks missing relevant information, while too many risks exceeding the LLM’s context window and introducing irrelevant content.

Persisting the index - The vector index is saved to disk via `StorageContext`, so it only needs to be built once across sessions rather than reprocessing all 145 documents every time.